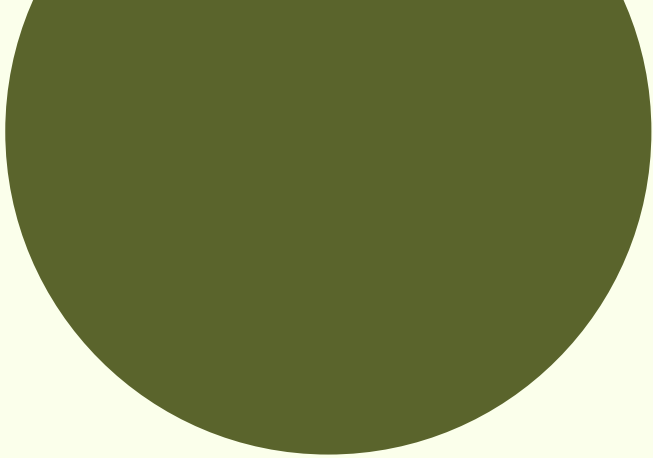




Automated Student Lab Allocation Using Maximum Bipartite Matching

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DSA 3
Section 7

* Abstract

- Laboratory allocation is a common scheduling challenge involving student preferences and limited lab capacities.
- Manual or naïve allocation can result in inefficient assignments, dissatisfaction, and resource underutilization.
- The proposed system models students and laboratories as a bipartite graph, where edges represent student preferences.
- Maximum Bipartite Matching will be used to maximize the number of successful student-lab assignments.
- The system will generate allocation reports and provide visualization of the matching process.
- The project demonstrates the practical application of graph matching algorithms in academic scheduling and resource allocation



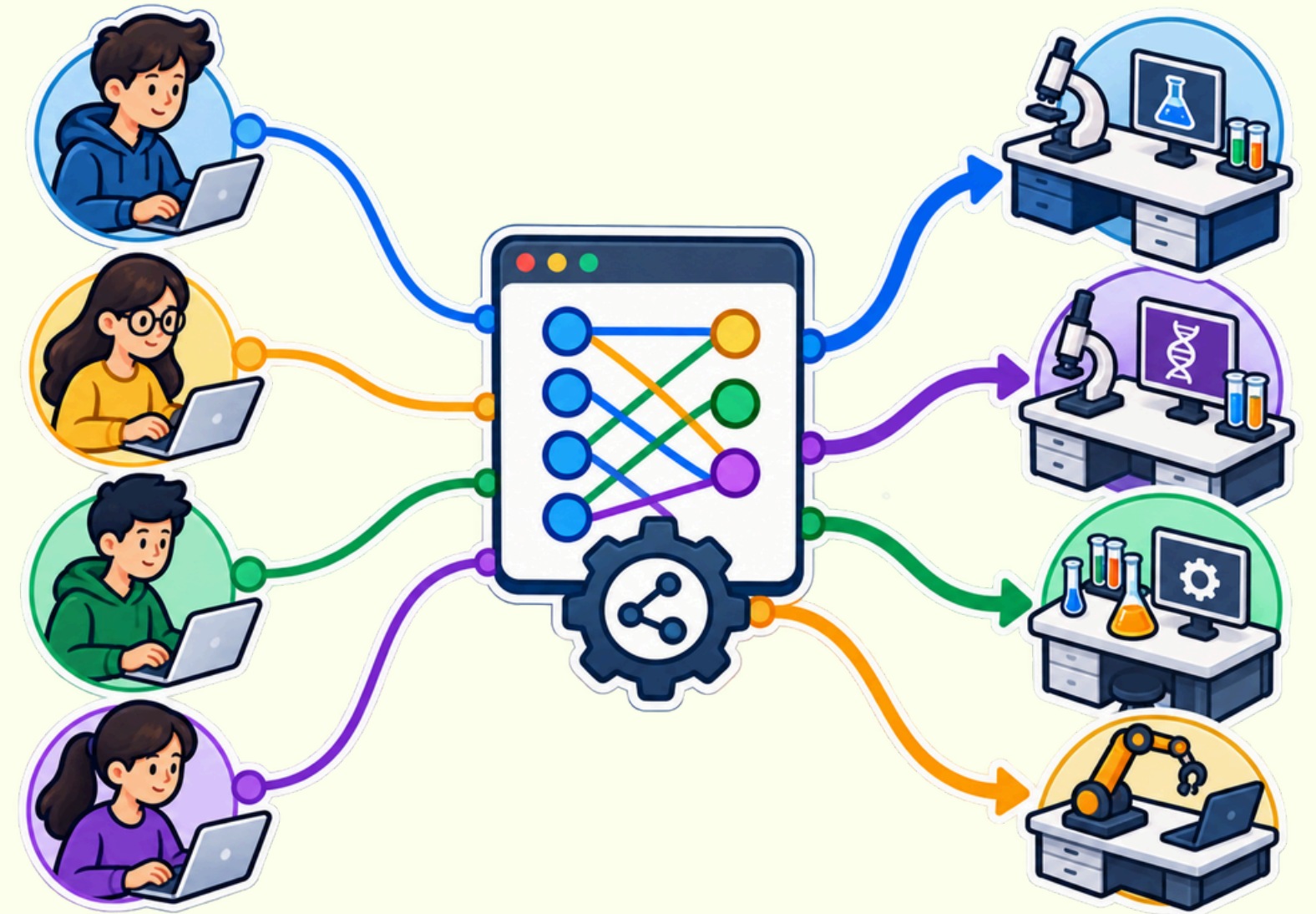
Problem Statement :

- *Assigning students to laboratory sessions based on preferences and capacity constraints is a common scheduling challenge.*
- *Manual or naïve allocation can lead to:*
 - *Inefficient assignments*
 - *Student dissatisfaction*
 - *Underutilization of lab resources*
- *The proposed system will automatically assign students to labs using Maximum Bipartite Matching.*
- *Each student will receive at most one lab allocation.*
- *The system will aim to maximize the total number of successful assignments.*
- *It will also provide allocation reports and visualization of the matching process.*



Proposed * Functionalities *

- **Student Management** – Enter student details and preferences.
- **Lab Management** – Add laboratories and define their capacities.
- **Bipartite Graph Creation** – Represent student-lab preferences as graph edges.
- **Maximum Bipartite Matching** – Find the maximum number of successful allocations.
- **Automatic Allocation** – Assign students to suitable labs while respecting constraints.
- **Allocation Reports** – Display student assignments and lab utilization.
- **Graph Visualization** – Visualize students, labs, and matched edges.



Tools and Technologies to be used:

Tools :

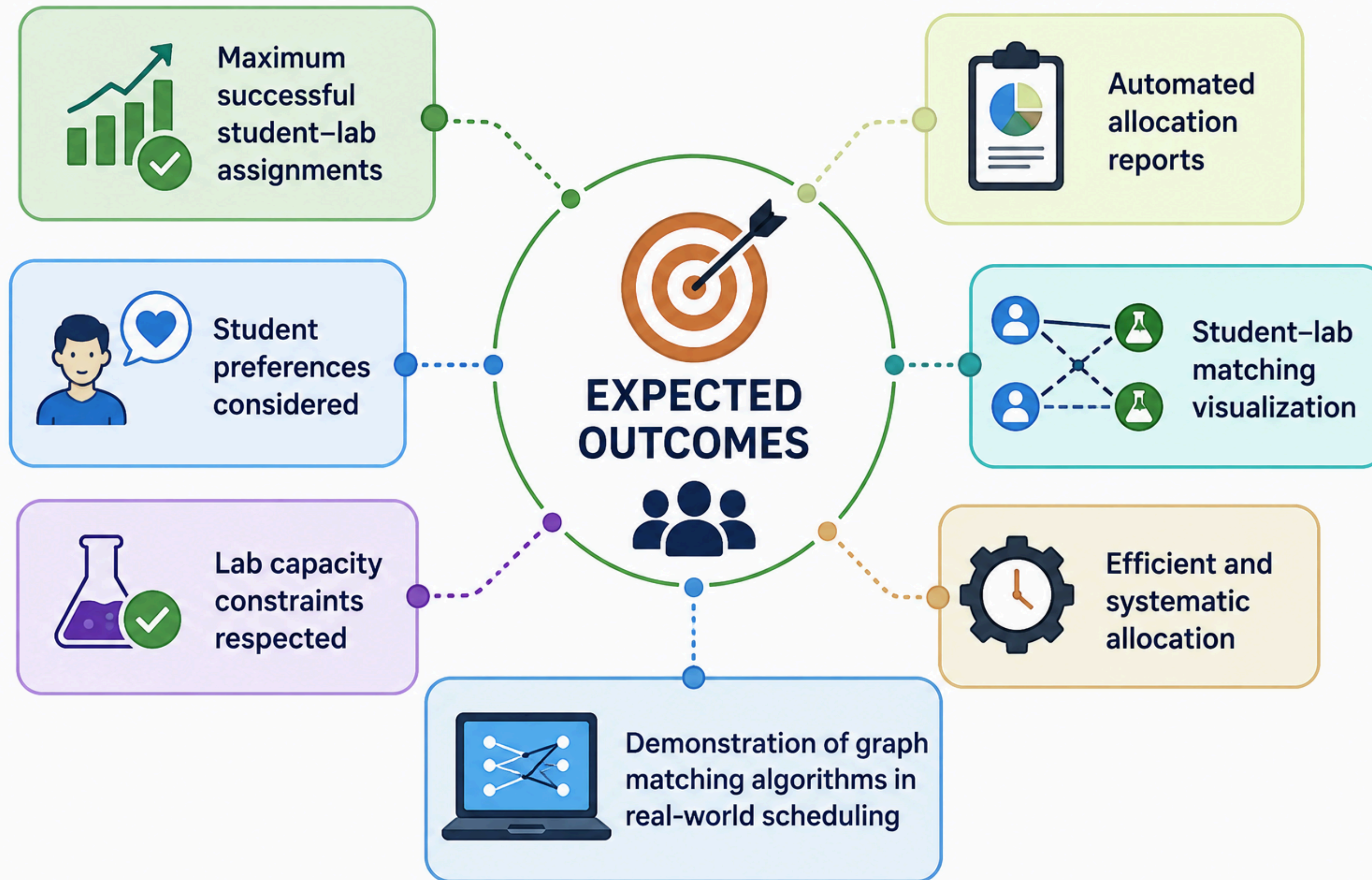
- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Java
- **Core Algorithm:** Maximum Bipartite Matching / Hopcroft-Karp
- **Data Structure:** Bipartite Graph
- **Database:** MySQL
- **Graph Visualization:** JavaScript-based visualization
- **Development Environment:** Visual Studio Code / IntelliJ IDEA
- **Version Control:** Git & GitHub

Core Technologies :

Graph Theory + Maximum Bipartite Matching + Database + Web Interface



Expected Outcomes :



Summary :



The Problem

Students have preferences for labs, but each lab has limited capacity.



S1

Prefers:
Lab A, Lab B



S2

Prefers:
Lab A, Lab C



S3

Prefers:
Lab B, Lab C



S4

Prefers:
Lab A, Lab B



Lab A

Capacity: 2



Lab B

Capacity: 1



Lab C

Capacity: 2



Manual allocation leads to:

- Inefficient assignments
- Student dissatisfaction
- Underutilized resources



Our Approach

Maximum Bipartite Matching

Students

Labs

S1

S2

S3

S4

A

B

C



Algorithm finds the maximum number of successful assignments while respecting:

- Student preferences
- Lab capacities



Result

Student

Assigned Lab

S1

Lab A



S2

Lab C



S3

Lab B



S4

Lab A



Outcome:

- ✓ Maximum assignments
- ✓ No student in multiple labs
- ✓ Lab capacities respected
- ✓ Allocation report generated
- ✓ Graph visualization provided



Thank You

Team 07